AUTOMATED SUSHI ASSEMBLY LINE

PROJECT PROPOSAL REPORT

Class: Capstone Design Project I Independent Study

Course Code: 4TR1 C01

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Project Statement

This project aims to design and build a fully automated sushi assembly line that is capable of producing simple, high-quality sushi rolls. It will integrate various technologies such as a conveyor belt, a robotic arm with a pick-and-place end effector, and processing stations meant to simulate the core steps of sushi preparation.

The assembly line can be broken down into the following stations:

- Assembling the rice and nori
- Laying down the fish/toppings
- Rolling everything together
- Cutting the roll
- Plating the pieces

Altogether, this project strives to make the process of assembling sushi more efficient and consistent, thereby increasing quality and lowering costs in turn. Primarily being used in restaurants, this assembly line will automate the menial task of creating basic sushi rolls, which allows the more skilled workers and trained chefs to focus on the creative, supervisory, and customer-facing roles.

Project Objectives

The following statements are high-level objectives for the automated sushi assembly line.

- It shall produce uniform pieces of sushi that are consistent in appearance and quality
- It shall not require human intervention to complete the process, aside from replenishing the ingredients/materials
- It shall be safe for humans to interact with the system if need be
- It shall be entirely food-grade and up to food safety requirements
- It shall have an HMI to display relevant data and allow for emergency control (ex. runtime data, emergency stops, etc.)
- It shall be quick and efficient, taking less time than it would for an average human to execute the process

Background

Throughout the BTech program, we have been exposed to a wide range of technologies and technical skills that prepare us to contribute to the ever-growing automation industry. Given the freedom, we wanted to select a human-centered project that we could design, build, and evaluate from beginning to end. Automating the assembly of sushi presents a familiar and relatable task with real-world applications, allowing people to easily see the value of our project in addressing repetitive manual processes. As the designers and end-users of our product, our motivation stems from its practicality and relevance to our own lives. The project's tangible outcome engages us and encourages us to take ownership of any technical or design challenges that we may face.

This project provides a great opportunity to enhance our capabilities as we integrate the skills that we have developed over the last four years into a comprehensive system that we designed from scratch. The sushi assembly process involves several stages, each with their own mechanical, control, and electrical requirements that will allow us to demonstrate our creativity and engineering judgement. Coordinating the different stations into a cohesive product is bound to present challenges on a technical and interpersonal front, in terms of managing systems integrations and team collaboration. But successfully completing this project will display our abilities in coding, electrical architecture, and mechanical designs, while also strengthening our experience in collaborative engineering and real-world project management.

Project Schedule

Week 1: Project idea generation and brainstorming

Week 2: Idea selection and preliminary design ideas

Week 3: Confirming project idea and generating project proposal

Week 4: Presenting proposal and developing project requirements

Week 5: Narrowing down design ideas according to desired specifications

Week 6: Developing design prototypes and testing

Week 7: Mid-term report and presentation

Week 8: Creating a bill of materials and sourcing parts (including software, and how to use those parts)

Week 9: Buy key parts for project that we will build around

Week 10: Build prototype with reduced functionality (ex. 1/5 stations in the process actually functioning)

Week 11: Create poster for project

Week 12: Put together the final report

Week 13: Final poster presentation and report submission